

Advanced Algorithms

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1 Approximation Algorithms

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Definition 1.1 (Problem).

$$\begin{aligned}\Pi &:= (\mathcal{I}, \mathcal{S}, \mathcal{F}). \\ \mathcal{I} &:= \{ I : I \text{ is an instance} \}, \quad \mathcal{S} := \{ S : S \text{ is a solution} \}, \\ \mathcal{F} &: \mathcal{I} \rightarrow 2^{\mathcal{S}} \text{ feasibility map.}\end{aligned}$$

Definition 1.2 (Optimization problem).

$$\begin{aligned}\Pi &:= (\mathcal{I}, \mathcal{S}, \mathcal{F}, c, \text{goal}). \\ \text{goal} &\in \{\min, \max\}, \quad c : \mathcal{I} \times \mathcal{S} \rightarrow \mathbb{R}_{\geq 0} \text{ cost function.} \\ I &\in \mathcal{I}, \quad \mathcal{F}(I) \neq \emptyset. \\ \text{OPT}(I) \in \mathcal{F}(I) : \quad c(I, \text{OPT}(I)) &= \begin{cases} \min_{S \in \mathcal{F}(I)} c(I, S), & \text{goal} = \min, \\ \max_{S \in \mathcal{F}(I)} c(I, S), & \text{goal} = \max. \end{cases}\end{aligned}$$

Definition 1.3 (Algorithm).

$$\begin{aligned}\mathcal{A} : \mathcal{I} &\rightarrow \mathcal{S}, \quad \mathcal{A}(I) \in \mathcal{F}(I). \\ \Pi &:= (\mathcal{I}, \mathcal{S}, \mathcal{F}) \text{ problem.}\end{aligned}$$

Definition 1.4 (Polynomial).

$$\text{poly}(n) := \{ f : \exists c \in \mathbb{N}, f \in O(n^c) \}.$$

Definition 1.5 (Running time (TODO)).

Definition 1.6 (Efficient algorithm).

$$\text{time}_{\mathcal{A}}(I) \leq \text{poly}(|I|).$$

Definition 1.7 (α -approximation).

$$\mathcal{A} \text{ is an } \alpha\text{-approximation} \iff \forall I \in \mathcal{I} : \begin{cases} c(I, \mathcal{A}(I)) \leq \alpha c(I, \text{OPT}(I)), & \text{goal} = \min, \\ c(I, \text{OPT}(I)) \leq \alpha c(I, \mathcal{A}(I)), & \text{goal} = \max. \end{cases}$$

$$\Pi := (\mathcal{I}, \mathcal{S}, \mathcal{F}, c, \text{goal}) \text{ optimization problem,} \quad \mathcal{A} : \mathcal{I} \rightarrow \mathcal{S}, \mathcal{A}(I) \in \mathcal{F}(I) \text{ algorithm,} \quad \alpha \in \mathbb{R}_{\geq 1}.$$